

Standardising IoT STANDARDS And GROUPS



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No matter which company or application area you talk about, the Internet of Things (IoT) seems to be the trending buzzword. One of the next big things surely, but it is also a very confusing domain. With many standards and overlapping member alliances, it is not an easy arena to work in. This article tries to provide some clarity on the standards aspect of the IoT.

Should you be worrying about this

“Without standards, the IoT will fail to realise the full potential predicted for it,” feels Martin Woolley, technical program manager at Bluetooth Special Interest Group (SIG). “It is said of teams

that they go through phases of forming, storming, norming and performing. I think something similar can be said to be happening with the IoT, though it may be difficult to achieve a consensus as to which of these earlier phases we are currently in,” he adds.

It is for all to see that major organisations are into multiple standards-framing associations, not just one. It just goes to show that we are in a highly-experimental phase and the winner is, as yet, unclear. The field is open and up for grabs, but we can be certain that the winner is going to be backed heavily.

The three major competitors

The three incumbents aiming to become the standard for tomorrow’s IoT are Alljoyn, IoTivity and Thread. Let us begin by trying to understand what each of these has to offer.

Alljoyn. Alljoyn is an open source effort by AllSeen Alliance, with the source code starting off at Qualcomm and currently with Linux Foundation. With more than 170 members including LG, Sharp, Haier, Panasonic, Sony, Electrolux, Sears, Silicon Image, Cisco, TP-Link, Canary, Two Bulls and Xiaomi, the group has based this standard on the client-server model where the server/producer advertises the device’s capability and the client accepts or ignores it.

The service framework behind Alljoyn aims at creating a rich user experience by automating how a device connects to this network. With the latest addition of a device system bridge by Microsoft, even devices using

